

Multiple Linear Regression

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- 2 Example Problem
- 3 Multiple Linear Regression with Continuous Explanatory Variables
- 4 Multiple Linear Regression with Categorical Explanatory Variables
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Simple versus Multiple Linear Regression

Population models

Simple Linear Regression

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

Where:

- i refers to a specific observation.
- β_0 is the intercept parameter.
- β_1 is the slope parameter.
- ϵ_i is the error .

There is one dependent variable and one independent variable

Multiple Linear regression

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \dots + \beta_p x_{pi} + \epsilon_i$$

Where:

- i refers to an observation.
- p is the number of independent variables.
- β_0 is the intercept parameter.
- β_j is the slope parameter for the j^{th} independent variable ($j = 1, \dots, p$).
- ϵ_i is the error.

There is one dependent variable but two or more independent variables.

Sample models

Simple Linear Regression

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

Where:

- i refers to a specific observation.
- $\hat{y}_i$ is the predicted value of the dependent variable.
- $\hat{\beta}_0$ is the intercept estimate.
- $\hat{\beta}_1$ is the slope estimate.

There is one dependent variable and one independent variable.

Multiple Linear regression

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{1i} + \hat{\beta}_2 x_{2i} + \dots + \hat{\beta}_p x_{pi}$$

Where:

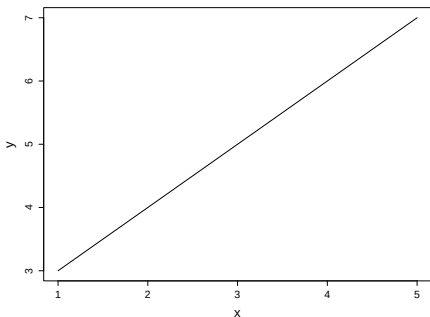
- i refers to an observation.
- p is the number of independent variables.
- $\hat{y}_i$ is the predicted value of the dependent variable.
- $\hat{\beta}_0$ is the intercept estimate.
- $\hat{\beta}_j$ is the slope estimate for the j^{th} independent variable ($j = 1, \dots, p$).

There is one dependent variable but two or more independent variables.

Visualisation of simple versus multiple regression

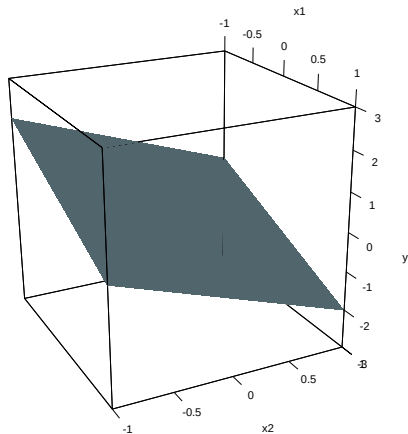
Simple linear regression model

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$



Multiple linear regression model

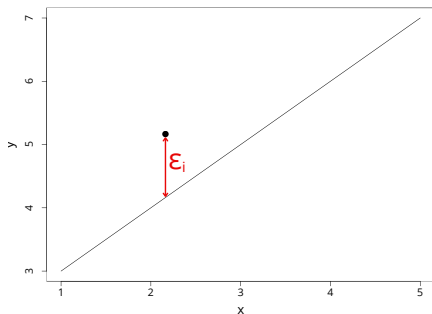
$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{1i} + \hat{\beta}_2 x_{2i}$$



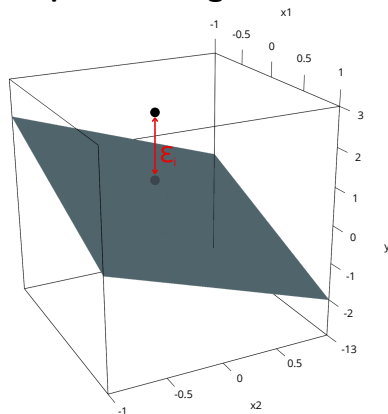
Estimating the β parameters

Like in simple linear regression, the Ordinary Least Squares (OLS) algorithm is used to estimate the β parameters by minimising $\sum_i \epsilon_i^2$.

Simple linear regression model



Multiple linear regression model



Aim of multiple regression

- Model a response (dependent) variable using many explanatory (independent) variables.
- Predict the value of a response variable from the values of multiple explanatory variables.
- Understand how a response variable changes with many different explanatory variables.

Example Problem

Fresh example

Company A produces Fresh, a brand of liquid laundry detergent. In order to manage its inventory more effectively and make revenue projections, this company would like to better predict the demand for Fresh. To develop a prediction model, the company has gathered data concerning demand for Fresh over the last 30 sales periods. The first few lines of this dataset are shown below:

```
> head(fresh)
  demand fresh_price ads_expenditure size ads_campaign competitor_price
1   7.38         3.85           5.50 Small          B             3.80
2   8.51         3.75           6.75  Big          B             4.00
3   9.52         3.70           7.25  Big          B             4.30
4   7.50         3.70           5.50 Small          A             3.70
5   9.33         3.60           7.00  Big          C             3.85
6   8.28         3.60           6.50 Small          A             3.80
```

- y (demand) = the demand for Fresh (100 000s of bottles).
- x_1 (fresh_price) = the price of Fresh detergent (in 10 rands).
- x_2 (ads_expenditure) = company's advertising expenditure to promote Fresh detergent (in 1000 rands).
- x_3 (size) = size of company (big or small).
- x_4 (ads_campaign) = company's advertising campaign (A: TV commercials, B: mixture of TV & radio, C: mixture of TV, radio, magazine & newspaper ads).
- x_5 (competitor_price) = average industry price of competitor's similar detergents (in 10 rands).

Correlation Analysis

Correlation analysis is done the same way for multiple x variables as it is for one. We just calculate the correlation between y and each x_j (so we have p correlation coefficients).

Often we also want to find the correlations between each x variable and every other x variable (It will become clear why shortly).

Note: we can do this analysis by hand but it would involve calculating many correlation coefficients so we instead use software like R.

Correlation analysis in R

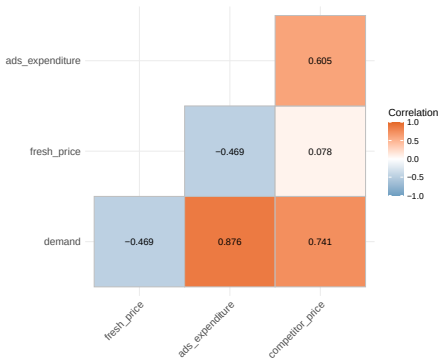
Because we want the correlations between all of our variables, a convenient way to do this is by calculating a correlation matrix. To do this in R we use:

```
fresh_numeric <- fresh[,c(1,2,3,6)]
cor(fresh_numeric)
```

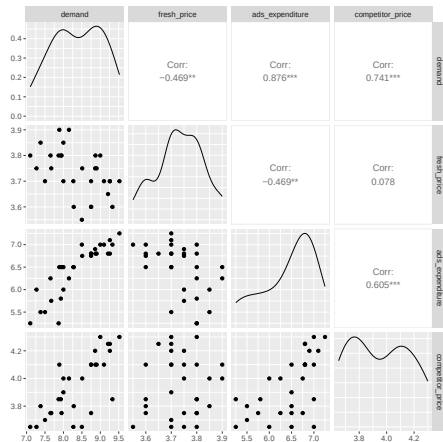
```
##                demand fresh_price ads_expenditure competitor_price
## demand           1.0000000 -0.4692203           0.8759536       0.74094817
## fresh_price      -0.4692203  1.0000000           -0.4687930       0.07836681
## ads_expenditure  0.8759536 -0.4687930           1.0000000       0.60454000
## competitor_price 0.7409482  0.07836681           0.6045400       1.00000000
```

- The diagonals represent the correlation between each variable and itself so these are always 1.
- You will also notice that below and above the diagonal is mirrored (because each variable occurs twice, once in the rows and once in columns).

Correlation Matrix Plot:



Pairwise Plot:



Correlations with dependent variable

```
##                demand fresh_price ads_expenditure competitor_price
## demand          1.0000000 -0.46922030      0.8759536      0.74094817
## fresh_price     -0.4692203  1.00000000      -0.4687930      0.07836681
## ads_expenditure  0.8759536 -0.46879301       1.0000000      0.60454000
## competitor_price 0.7409482  0.07836681       0.6045400      1.00000000
```

demand and fresh_price: $r = -0.469$

This indicates a negative moderate correlation between demand and fresh_price.

demand and ads_expenditure: $r = 0.876$

This indicates a positive strong correlation between demand and ads_expenditure.

demand and competitor_price: $r = 0.741$

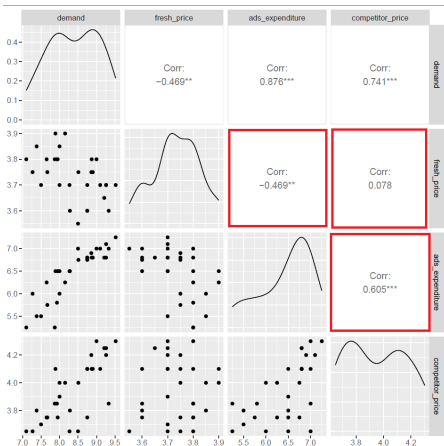
This indicates a positive quite strong correlation between demand and competitor_price.

Multicollinearity

Multicollinearity refers to high correlations between independent variables in a regression analysis. This means that the correlated variables would provide redundance information to the regression analysis, as they are attempting to explain the same variability in the dependent variable.

Assessing multicollinearity in a dataset is why we would also be interested in assessing the correlations between the independent variables in a correlation analysis.

```
##          demand fresh_price ads_expenditure competitor_price
## demand      1.0000000 -0.46922030      0.8759536      0.74094817
## fresh_price  -0.4692203  1.00000000     -0.4687930     0.07836681
## ads_expenditure 0.8759536 -0.46879301     1.0000000     0.60454000
## competitor_price 0.7409482  0.07836681     0.6045400     1.00000000
```



- There is no concern regarding the correlation between competitor_price and fresh_price as the correlation is very weak ($r = 0.078$).
- There is some concern regarding the correlation between ads_expenditure and fresh_price as there is a moderate correlation between these variables ($r = -0.469$).
- There is concern regarding the correlation between competitor_price and ads_expenditure as there is a relatively strong correlation between these variables ($r = 0.605$).

Consequences of multicollinearity for a regression analysis

When two independent variables are correlated in a regression analysis this means that both variables are trying to explain the same variability in the dependent variable. There is no way to determine which variable should get to explain that variability and this causes uncertainty in the parameter estimates. The result is:

- Inflated standard errors.
- Inflated p-values as a consequence of the inflated standard errors.
- Large changes to parameter estimates when removing or adding a correlated independent variable from or to the model.
- An increase in the RSE when adding a correlated independent variable to the model.

Multiple Linear Regression with Continuous Explanatory Variables

Performing Multiple Linear Regression in R

```
fit <- lm(demand ~ fresh_price + ads_expenditure + competitor_price, data = fresh)
summary(fit)
```

```
##
## Call:
## lm(formula = demand ~ fresh_price + ads_expenditure + competitor_price,
##     data = fresh)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.41065 -0.11562 -0.00984  0.13466  0.51361
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    7.5891     2.4450   3.104 0.004567 **
## fresh_price   -2.3577     0.6379  -3.696 0.001028 **
## ads_expenditure  0.5012     0.1259   3.981 0.000491 ***
## competitor_price 1.6122     0.2954   5.459 1.01e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2347 on 26 degrees of freedom
## Multiple R-squared:  0.8936, Adjusted R-squared:  0.8813
## F-statistic: 72.8 on 3 and 26 DF, p-value: 8.883e-13
```

- $\beta_0 \longrightarrow$ (Intercept) estimate
- $\beta_1 \longrightarrow$ fresh_price estimate
- $\beta_2 \longrightarrow$ ads_expenditure estimate
- $\beta_3 \longrightarrow$ competitor_price estimate

So the regression equation for the example is given by:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1(\text{fresh_price}) + \hat{\beta}_2(\text{ads_expenditure}) + \hat{\beta}_3(\text{competitor_price})$$

$$\hat{y} = 7.5891 - 2.3577(\text{fresh_price}) + 0.5012(\text{ads_exp}) + 1.6122(\text{comp_price})$$

Interpretation of β coefficients

$$\hat{y} = 7.5891 - 2.3577(\text{fresh_price}) + 0.5012(\text{ads_exp}) + 1.6122(\text{comp_price})$$

For our example:

Interpretation of β_0 (intercept) On average, the demand for Fresh is $(7.5891 \times 100000) = 758910$ bottles when all independent variables take on a value of 0. (these values would never practically take on a value of zero so this doesn't really offer valuable insight.)

Interpretation of β_1 (slope) On average, the demand for Fresh decreases by $(2.3577 \times 100000) = 235770$ bottles for each additional R10 increase in the price of Fresh, holding all else constant.

Interpretation of β_2 (slope) On average, the demand for Fresh increases by $(0.5012 \times 100000) = 50120$ bottles for each additional R1000 increase in advertising expenditure, holding all else constant.

Interpretation of β_3 (slope) On average, the demand for Fresh increases by $(1.6122 \times 100000) = 161220$ bottles for each additional R10 increase in the average industry price of competitor's similar detergents, holding all else constant.

Interpretation of β coefficients

In general:

Interpretation of β_0 (intercept) It is the average estimated value of y when all $x_j = 0$ ($j = 1, \dots, p$).

Interpretation of β_j (slope) It is the average estimated change in y for a unit increase in x_j , holding all other independent variables constant. If β_j is positive then the change is an increase and negative means a decrease (It is important to be **specific**).

Testing the significance of the β_j estimates

Like for simple linear regression, we are interested in determining whether the slope estimates are significant so that we can make statistical inference. We can do a hypothesis test on these estimates as follows (we do a separate test for each β_j where $j = 1, \dots, p$):

Step 1: Define the null hypothesis $H_0: \beta_j = 0$

Step 2: Define the alternative hypothesis $H_a: \beta_j \neq 0$

Step 3: Define the significance level $\alpha = 0.05$

Step 4: Calculate the test statistic

$$tstat = \frac{\hat{\beta}_j - \beta_j}{se(\hat{\beta}_j)} = \frac{\hat{\beta}_j}{se(\hat{\beta}_j)} \sim t_{n-p-1}$$

Example using β_2 ($\beta_{\text{ads_expenditure}}$):

$$tstat = \frac{0.5012}{0.1259} = 3.981 \sim t_{26}$$

Step 5: Find the p-value

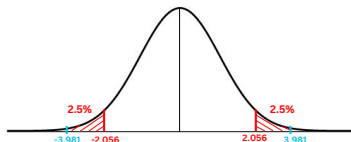


TABLE 2. *t*-DISTRIBUTION: One sided critical values, i.e. the value of t_{df}^P such that $P = \Pr[t_{df} > t_{df}^P]$, where df is the degrees of freedom

df	Probability Level P										
	0.4	0.3	0.2	0.1	0.05	0.025	0.01	0.005	0.0025	0.001	0.0005
1	0.325	0.727	1.376	3.078	6.314	12.71	31.82	63.66	127.3	318.3	636.6
2	0.289	0.617	1.061	1.886	2.920	4.303	6.965	9.925	14.09	22.33	31.60
3	0.277	0.584	0.978	1.638	2.353	3.182	4.541	5.841	7.453	10.21	12.92
4	0.271	0.569	0.941	1.533	2.132	2.776	3.747	4.604	5.598	7.173	8.610
5	0.267	0.559	0.920	1.476	2.015	2.571	3.365	4.032	4.773	5.894	6.869
6	0.265	0.553	0.906	1.440	1.943	2.447	3.143	3.707	4.317	5.208	5.959
7	0.263	0.549	0.896	1.415	1.895	2.365	2.998	3.499	4.029	4.785	5.408
8	0.262	0.546	0.889	1.397	1.860	2.306	2.896	3.355	3.833	4.501	5.041
9	0.261	0.543	0.883	1.383	1.833	2.262	2.821	3.250	3.690	4.297	4.781
10	0.260	0.542	0.879	1.372	1.812	2.228	2.764	3.169	3.581	4.144	4.587
11	0.260	0.540	0.876	1.363	1.796	2.201	2.718	3.106	3.497	4.025	4.437
12	0.259	0.539	0.873	1.356	1.782	2.179	2.681	3.055	3.428	3.930	4.318
13	0.259	0.538	0.870	1.350	1.771	2.160	2.650	3.012	3.372	3.852	4.221
14	0.258	0.537	0.868	1.345	1.761	2.145	2.624	2.977	3.326	3.787	4.140
15	0.258	0.536	0.866	1.341	1.753	2.131	2.602	2.947	3.286	3.733	4.073
16	0.258	0.535	0.865	1.337	1.746	2.120	2.583	2.921	3.252	3.686	4.015
17	0.257	0.534	0.863	1.333	1.740	2.110	2.567	2.898	3.222	3.646	3.965
18	0.257	0.534	0.862	1.330	1.734	2.101	2.552	2.878	3.197	3.610	3.922
19	0.257	0.533	0.861	1.328	1.729	2.093	2.539	2.861	3.174	3.579	3.883
20	0.257	0.533	0.860	1.325	1.725	2.086	2.528	2.845	3.153	3.552	3.850
21	0.257	0.532	0.859	1.323	1.721	2.080	2.518	2.831	3.135	3.527	3.819
22	0.256	0.532	0.858	1.321	1.717	2.074	2.508	2.819	3.119	3.505	3.792
23	0.256	0.532	0.858	1.319	1.714	2.069	2.500	2.807	3.104	3.485	3.768
24	0.256	0.531	0.857	1.318	1.711	2.064	2.492	2.797	3.091	3.467	3.745
25	0.256	0.531	0.856	1.316	1.708	2.060	2.485	2.787	3.078	3.450	3.725
26	0.256	0.531	0.856	1.315	1.706	2.056	2.479	2.779	3.067	3.435	3.707

```
fit <- lm(demand ~ fresh_price + ads_expenditure + competitor_price, data = fresh)
summary(fit)
```

```
##
## Call:
## lm(formula = demand ~ fresh_price + ads_expenditure + competitor_price,
##     data = fresh)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.41065 -0.11562 -0.00984  0.13466  0.51361
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    7.5891     2.4450   3.104 0.004567 **
## fresh_price    -2.3577     0.6379  -3.696 0.001028 **
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##
## Residual standard error: 0.2347 on 26 degrees of freedom
## Multiple R-squared:  0.8936, Adjusted R-squared:  0.8813
## F-statistic: 72.8 on 3 and 26 DF,  p-value: 8.883e-13
```

Step 6: Make a conclusion the p-value is sufficiently small to reject H_0 (p-value < 0.05). Thus, it can be concluded that there is some significant linear relationship between advertising expenditure and the demand for Fresh.

Confidence intervals for β_j estimates

We can get a range of values that we are relatively sure that the true β_j parameters fall into (with a given level of confidence) for each β_j .

We calculate this range using:

$$CI = \hat{\beta}_j \pm t_{\frac{\alpha}{2}, n-p-1} \times se(\hat{\beta}_j)$$

We can calculate the 95% confidence interval for β_2 ($\beta_{\text{ads_expenditure}}$) as follows:

$$CI = \hat{\beta}_2 \pm t_{\frac{\alpha}{2}, n-p-1} \times se(\hat{\beta}_2)$$

$$CI = 0.5012 \pm 2.056 \times 0.1259$$

$$CI = [0.242, 0.760]$$

```
confint(fit)
```

##		2.5 %	97.5 %
##	(Intercept)	2.5632892	12.6149158
##	fresh_price	-3.6690418	-1.0464038
##	ads_expenditure	0.2424145	0.7598889
##	competitor_price	1.0051083	2.2193205

Checking overall model significance

Like with simple linear regression, we can check the overall model significance by assessing if our model is significantly different to a null model (a model with just an intercept).

Source of Variation	df	SS	Mean Square	F
Regression	p	$SS_{\text{reg}} = \sum_i (\hat{y}_i - \bar{y})^2$	$MS_{\text{reg}} = \frac{SS_{\text{reg}}}{p}$	$\frac{MS_{\text{reg}}}{MS_{\text{resid}}}$
Residual	$n-p-1$	$SS_{\text{resid}} = \sum_i (y_i - \hat{y}_i)^2$	$MS_{\text{resid}} = \frac{SS_{\text{resid}}}{n-p-1}$	
Total	$n-1$	$SS_{\text{tot}} = \sum_i (y_i - \bar{y})^2$		

We can use the F-statistic to then perform an F-test.

Note: $\sqrt{MS_{\text{resid}}} = RSE$

Testing the overall model significance

Step 1: Define the null hypothesis $H_0: \beta_1 = \beta_2 = \dots = \beta_p = 0$

Step 2: Define the alternative hypothesis H_a : at least one $\beta_j \neq 0$

Step 3: Define the significance level $\alpha = 0.05$

Step 4: Calculate the test statistic

$$Fstat = \frac{\frac{SS_{reg}}{p}}{\frac{SS_{resid}}{n-p-1}} = \frac{MS_{reg}}{MS_{resid}} \sim F_{p, n-p-1}$$

Step 5: Find the p-value

```
fit <- lm(demand ~ fresh_price + ads_expenditure + competitor_price, data = fresh)
summary(fit)

##
## Call:
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```

Step 6: Make a conclusion the p-value is sufficiently small to reject H_0 (p-value < 0.05). Thus it can be concluded that there is a linear relationship between at least one of the independent variables and the demand for Fresh (ie. the overall model is significant).

Coefficient of Determination (R^2)

The coefficient of determination is calculated in exactly the same way as with simple linear regression:

$$R^2 = \frac{SS_{\text{reg}}}{SS_{\text{tot}}} = \frac{\sum_i (\hat{y}_i - \bar{y})^2}{\sum_i (y_i - \bar{y})^2}$$

We can see from looking at this equation that SS_{tot} will remain the same regardless of the model and SS_{reg} can only remain the same or increase when adding new variables to the model. This means that the value of R^2 never decreases when adding additional independent variables to the model.

This is a problem when using R^2 to compare models as there is a cost to the inclusion of additional variables. It means we lose degrees of freedom on our test statistics. This can lead to reduced power to detect a significant effect (power is the probability that a test of significance will pick up on an effect that does exist).

Adjusted coefficient of determination

To combat these limitations of the coefficient of determination, we introduce an adjusted version. This adjusts the R^2 value to consider sample size and the number of independent variables in the model and ensures it is still scaled between 0 and 1. The Adjusted R^2 is given by:

$$\text{Adjusted } R^2 = 1 - \left(\frac{n - 1}{n - p - 1} \right) (1 - R^2)$$

It now describes proportion of variation in the dependent variable as explained by all of the independent variables, taking sample size and number of independent variables into account.

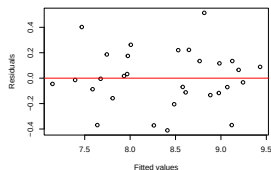
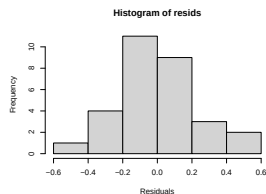
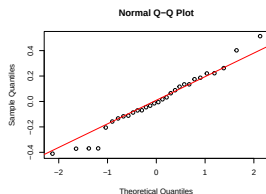
Adjusted coefficient of determination in R

```
fit <- lm(demand ~ fresh_price + ads_expenditure + competitor_price, data = fresh)
summary(fit)
```

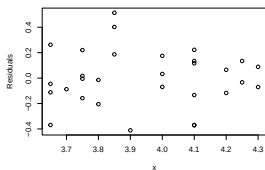
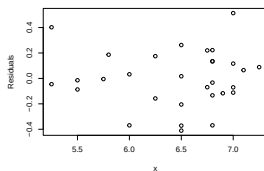
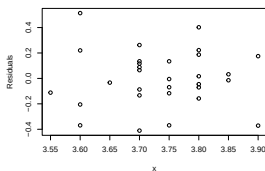
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##
## Call:
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## -0.41065 -0.11562 -0.00984  0.13466  0.51361
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    7.5891     2.4450   3.104 0.004567 **
## fresh_price    -2.3577     0.6379  -3.696 0.001028 **
## ads_expenditure  0.5012     0.1259   3.981 0.000491 ***
## competitor_price 1.6122     0.2954   5.459 1.01e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2347 on 26 degrees of freedom
## Multiple R-squared:  0.8936, Adjusted R-squared:  0.8813
## F-statistic: 72.8 on 3 and 26 DF, p-value: 8.883e-13
```

Model Checking

We can do the same model checks that we did for the simple linear regression section. Here are the plots for our example model:



For the plot of residuals versus independent variables we now just need to produce p plots instead of only 1 (where p is the number of independent variables).



Prediction

Prediction works exactly the same as it does for simple linear regression but now we just need to supply values for all of the independent variables to get a prediction for the dependent variable.

For our example the fitted line is given by:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1(\text{fresh_price}) + \hat{\beta}_2(\text{ads_expenditure}) + \hat{\beta}_3(\text{competitor_price})$$
$$\hat{y} = 7.5891 - 2.3577(\text{fresh_price}) + 0.5012(\text{ads_exp}) + 1.6122(\text{comp_price})$$

If we want to know the predicted demand for Fresh when the price of Fresh is R50, the average competitor price is R55 and advertising expenditure is R8000 we can do the following:

$$\hat{y} = 7.5891 - 2.3577(5) + 0.5012(8) + 1.6122(5.5)$$
$$\hat{y} = 8.6773$$

So the predicted demand for Fresh is 867730 bottles.

Confidence and Prediction interval

We can calculate prediction and confidence intervals for our predictions.

CI for average y:

PI for individual y:

$$\hat{y} \pm t_{\frac{\alpha}{2}, n-p} s_{\epsilon} \sqrt{x_i' (X'X)^{-1} x_i}$$

$$\hat{y} \pm t_{\frac{\alpha}{2}, n-p} s_{\epsilon} \sqrt{1 + x_i' (X'X)^{-1} x_i}$$

You do not need to know the formulas for these or how to use them but know how to get the intervals in R.

Prediction in R

```
#-----
#Predicting for a single individual
#-----

newdat <- list(fresh_price=5,ads_expenditure=8,competitor_price=5.5)
newdat <- data.frame(fresh_price=5,ads_expenditure=8,competitor_price=5.5)
predict(fit, newdata = newdat, interval = "confidence")
```

```
##          fit          lwr          upr
## 1 8.676881 7.091324 10.26244
```

```
predict(fit, newdata = newdat, interval = "prediction")
```

```
##          fit          lwr          upr
## 1 8.676881 7.019572 10.33419
```

Multiple Linear Regression with Categorical Explanatory Variables

Categorical explanatory variables

Sometimes we have datasets that do not only contain continuous variables. We previously just removed the categorical variables from our example dataset, but these categorical variables could also explain important variation in our dependent variable.

```
> head(fresh)
```

	demand	fresh_price	ads_expenditure	size	ads_campaign	competitor_price
1	7.38	3.85	5.50	Small	B	3.80
2	8.51	3.75	6.75	Big	B	4.00
3	9.52	3.70	7.25	Big	B	4.30
4	7.50	3.70	5.50	Small	A	3.70
5	9.33	3.60	7.00	Big	C	3.85
6	8.28	3.60	6.50	Small	A	3.80

For example, company size and the type of advertising campaign used could be important in determining the demand for Fresh.

How to include binary variables in a regression analysis

We will start by considering categorical variables with 2 levels. For our example dataset we have the variable size which indicates whether a company is small or big.

We can let Small take on the value 0 and Big take on the value 1, which gives us:

	fresh.size	size
1	Small	0
2	Big	1
3	Big	1
4	Small	0
5	Big	1
6	Small	0

This allows us to treat the categorical variable as numeric while still preserving the fact that it has 2 levels (0 or 1).

We can express the regression equation as before:

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{1i} + \hat{\beta}_2 x_{2i} + \hat{\beta}_3 x_{3i} + \hat{\beta}_4 x_{4i}$$

but now $\hat{\beta}_4$ represents the change in demand when a company is big versus small (ie. we multiply $\hat{\beta}_4$ by 0 when the company is small and 1 when the company is big).

What this essentially does is it treats Small as a reference category, so the intercept estimate captures the effect of Small on demand. When a company is Big we adjust the demand by $\hat{\beta}_4$.

How to include binary variables in a regression analysis in R

Default reference category:

By default when converting a character variable to a factor in R it uses the category that is first alphabetically as the reference category.

```
fresh$size <- as.factor(fresh$size)
str(fresh)
```

```
## 'data.frame': 30 obs. of 6 variables:
## $ demand : num 7.38 8.51 9.52 7.5 9.33 8.28 8.75 7.87 7.1 8 ...
## $ fresh_price : num 3.85 3.75 3.7 3.7 3.6 3.6 3.6 3.8 3.8 3.85 ...
## $ ads_expenditure : num 5.5 6.75 7.25 5.5 7 6.5 6.75 5.25 5.25 6 ...
## $ size : Factor w/ 2 levels "Big","Small": 2 1 1 2 1 2 1 2 2 2 ...
## $ ads_campaign : chr "B" "B" "B" "A" ...
## $ competitor_price: num 3.8 4 4.3 3.7 3.85 3.8 3.75 3.85 3.65 4 ...
```

```
contrasts(fresh$size)
```

```
##      Small
## Big      0
## Small    1
```

Manual reference category: We can also create our own new variable that uses the other category as the reference.

```
size_Big <- ifelse(fresh$size=="Small",0,1)
size_Big <- factor(size_Big, labels = c("Small","Big"))
contrasts(size_Big)
```

```
##      Big
## Small  0
## Big    1
```

```
fresh$size_Big <- size_Big
str(fresh)
```

```
## 'data.frame': 30 obs. of 7 variables:
## $ demand : num 7.38 8.51 9.52 7.5 9.33 8.28 8.75 7.87 7.1 8 ...
## $ fresh_price : num 3.85 3.75 3.7 3.7 3.6 3.6 3.6 3.8 3.8 3.85 ...
## $ ads_expenditure : num 5.5 6.75 7.25 5.5 7 6.5 6.75 5.25 5.25 6 ...
## $ size : Factor w/ 2 levels "Big","Small": 2 1 1 2 1 2 1 2 2 2 ...
## $ ads_campaign : chr "B" "B" "B" "A" ...
## $ competitor_price: num 3.8 4 4.3 3.7 3.85 3.8 3.75 3.85 3.65 4 ...
## $ size_Big : Factor w/ 2 levels "Small","Big": 1 2 2 1 2 1 2 1 1 1 ...
```

How to include categorical variables in a regression analysis

When a categorical variable has more than 2 levels, we do the same thing but now we just have two dummy variables instead of one. This also means 2 β estimates (one for each dummy variable).

If we now consider adding `ads_campaign` to our model with size (the companies advertising campaign: A, B or C), this would add two additional β estimates to the model ($\hat{\beta}_5$ and $\hat{\beta}_6$)

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{1i} + \hat{\beta}_2 x_{2i} + \hat{\beta}_3 x_{3i} + \hat{\beta}_4 x_{4i} + \hat{\beta}_5 x_{5i} + \hat{\beta}_6 x_{6i}$$

Where $\hat{\beta}_5$ would be the change in demand for fresh when advertising campaign is campaign B compared to campaign A and $\hat{\beta}_6$ would be the change in demand for fresh when advertising campaign is campaign C compared to campaign A (assuming we used A as the reference category).

How to include categorical variables in a regression analysis in R

Default reference category: We can use the default the same as with a binary variable and this will use the category alphabetically first as the reference.

```
fresh$ads_campaign <- as.factor(fresh$ads_campaign)
str(fresh)

## 'data.frame': 30 obs. of 7 variables:
## $ demand : num 7.38 8.51 9.52 7.5 9.33 8.28 8.75 7.87 7.1 8 ...
## $ fresh_price : num 3.85 3.75 3.7 3.7 3.6 3.6 3.6 3.8 3.8 3.85 ...
## $ ads_expenditure : num 5.5 6.75 7.25 5.5 7 6.5 6.75 5.25 5.25 6 ...
## $ size : Factor w/ 2 levels "Big","Small": 2 1 1 2 1 2 1 2 2 2 ...
## $ ads_campaign : Factor w/ 3 levels "A","B","C": 2 2 2 1 3 1 3 3 2 3 ...
## $ competitor_price: num 3.8 4 4.3 3.7 3.85 3.8 3.75 3.85 3.65 4 ...
## $ size_Big : Factor w/ 2 levels "Small","Big": 1 2 2 1 2 1 2 1 1 1 ...

contrasts(fresh$ads_campaign)

## B C
## A 0 0
## B 1 0
## C 0 1
```

Manual reference category: We can create 2 sets of dummy variables which means that the category that is not included as a dummy will be the reference.

```
campaignA <- ifelse(fresh$ads_campaign=="A",1,0)
campaignC <- ifelse(fresh$ads_campaign=="C",1,0)
campaignA <- factor(campaignA, labels = c("notA","A"))
campaignC <- factor(campaignC, labels = c("notC","C"))

fresh$campaignA <- campaignA
fresh$campaignC <- campaignC
str(fresh)

## 'data.frame': 30 obs. of 9 variables:
## $ demand : num 7.38 8.51 9.52 7.5 9.33 8.28 8.75 7.87 7.1 8 ...
## $ fresh_price : num 3.85 3.75 3.7 3.7 3.6 3.6 3.6 3.8 3.8 3.85 ...
## $ ads_expenditure : num 5.5 6.75 7.25 5.5 7 6.5 6.75 5.25 5.25 6 ...
## $ size : Factor w/ 2 levels "Big","Small": 2 1 1 2 1 2 1 2 2 2 ...
## $ ads_campaign : Factor w/ 3 levels "A","B","C": 2 2 2 1 3 1 3 3 2 3 ...
## $ competitor_price: num 3.8 4 4.3 3.7 3.85 3.8 3.75 3.85 3.65 4 ...
## $ size_Big : Factor w/ 2 levels "Small","Big": 1 2 2 1 2 1 2 1 1 1 ...
## $ campaignA : Factor w/ 2 levels "notA","A": 1 1 1 2 1 2 1 1 1 1 ...
## $ campaignC : Factor w/ 2 levels "notC","C": 1 1 1 1 2 1 2 2 1 2 ...
```

Regression analysis using our example in R

```
fit_full <- lm(demand ~ fresh_price + ads_expenditure + competitor_price + size +
              ads_campaign, data = fresh))
summary(fit_full)

##
## Call:
## lm(formula = lm(demand ~ fresh_price + ads_expenditure + competitor_price +
##   size + ads_campaign, data = fresh))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.30097 -0.09306  0.01320  0.09232  0.25139
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    8.29737     1.55158   5.348 1.98e-05 ***
## fresh_price    -2.31029     0.48687  -4.745 8.78e-05 ***
## ads_expenditure  0.43259     0.08599   5.031 4.32e-05 ***
## competitor_price 1.46893     0.22020   6.671 8.34e-07 ***
## sizeSmall     -0.19425     0.11761  -1.652 0.11221
## ads_campaignB   0.22263     0.07283   3.057 0.00559 **
## ads_campaignC   0.41478     0.06957   5.962 4.44e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1451 on 23 degrees of freedom
## Multiple R-squared:  0.964, Adjusted R-squared:  0.9546
## F-statistic: 102.6 on 6 and 23 DF,  p-value: 1.972e-15
```

- $\beta_0 \longrightarrow$ (Intercept) estimate
- $\beta_1 \longrightarrow$ fresh_price estimate
- $\beta_2 \longrightarrow$ ads_expenditure estimate
- $\beta_3 \longrightarrow$ competitor_price estimate
- $\beta_4 \longrightarrow$ sizeSmall estimate
- $\beta_5 \longrightarrow$ ads_campaign B estimate
- $\beta_6 \longrightarrow$ ads_campaign C estimate

So the regression equation for the example is given by:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1(\text{fresh_price}) + \hat{\beta}_2(\text{ads_expenditure}) + \hat{\beta}_3(\text{competitor_price}) + \hat{\beta}_4(\text{sizeSmall}) + \hat{\beta}_5(\text{ads_campaignB}) + \hat{\beta}_6(\text{ads_campaignC})$$

$$\hat{y} = 8.297 - 2.310(\text{fresh_price}) + 0.433(\text{ads_exp}) + 1.469(\text{comp_price}) - 0.194(\text{sizeSmall}) + 0.223(\text{ads_campaignB}) + 0.415(\text{ads_campaignC})$$

Interpretation of β coefficients for categorical variables

$$\hat{y} = 8.297 - 2.310(\text{fresh_price}) + 0.433(\text{ads_exp}) + 1.469(\text{comp_price}) - 0.194(\text{sizeSmall}) + 0.223(\text{ads_campaignB}) + 0.415(\text{ads_campaignC})$$

For our example:

Interpretation of β_0 (Intercept) On average, the demand for Fresh is $(8.29737 \times 100000) = 829737$ bottles when the price of Fresh is R0, the advertisement expenditure is R0, the competitor price is R0, the company size is big and the advertising campaign is A.

Interpretation of β_4 On average, the demand for Fresh is $(0.19425 \times 100000) = 19425$ bottles lower when the company is small compared to big, while holding all else constant.

Interpretation of β_5 On average, the demand for Fresh is $(0.22263 \times 100000) = 22263$ bottles higher for a company using advertising campaign B compared to a company using advertising campaign A, while holding all else constant.

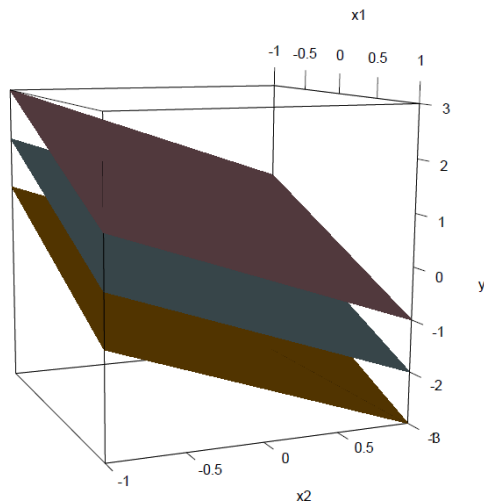
Interpretation of β_6 On average, the demand for Fresh is $(0.41478 \times 100000) = 41478$ bottles higher for a company using advertising campaign C compared to a company using advertising campaign A, while holding all else constant.

Interpretation of β coefficients for categorical variables

In general:

- Intercept** It is the average estimated value of y when all continuous independent variables=0 and categorical independent variables are at the reference category.
- $\beta_{\text{category}_i}$ It is the average estimated change in y while $x_{\text{categorical}}$ takes on dummy category i compared to when $x_{\text{categorical}}$ takes on the reference category, holding all else constant.

Visual demonstration of the change in intercept for different levels of categorical variable



Testing significance of categorical β_j estimates

The test is exactly the same as it is for continuous β_j estimates, the only difference is the conclusion of the test:

If H_0 is rejected The p-value is sufficiently small to reject H_0 (p-value < significance level). Thus, it can be concluded that there is a significant difference in y between *dummy category* and *reference category*.

If H_0 is not rejected The p-value is not sufficiently small to reject H_0 (p-value > significance level). Thus, no significant difference in y was detected between *dummy category* and *reference category*.

Interactions

What we have been looking at up until this point are additive regression models, these are models where the response is modelled by some scaled sum of the independent variables, for example:

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{1i} + \hat{\beta}_2 x_{2i}$$

But sometimes we may suspect that the relationship between y and x_1 is affected by the value of x_2 . This means that we want to modify the effect of x_1 on y for different values of x_2 . We can do this by introducing an interaction term to our regression model. This results in a multiplicative model rather than an additive one:

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{1i} + \hat{\beta}_2 x_{2i} + \hat{\beta}_3 x_{1i} x_{2i}$$

Now the effect of x_1 on y is given by: $(\hat{\beta}_1 + \hat{\beta}_3 x_2)$ not just $\hat{\beta}_1$.

Adding interaction terms to regression models in R

We use the * instead of + in the regression equation to represent an interaction in R.

```
fit <- lm(demand ~ fresh_price * ads_campaign, data = fresh)
summary(fit)

##
## Call:
## lm(formula = demand ~ fresh_price * ads_campaign, data = fresh)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.90983 -0.30334 -0.02153  0.42799  0.92383
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      12.226      6.548   1.867  0.0741 .
## fresh_price       -1.079      1.766  -0.611  0.5471
## ads_campaignB      35.186     12.836   2.741  0.0114 *
## ads_campaignC      13.538      9.435   1.435  0.1642
## fresh_price:ads_campaignB  -9.307      3.433  -2.711  0.0122 *
## fresh_price:ads_campaignC  -3.549      2.531  -1.402  0.1737
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5507 on 24 degrees of freedom
## Multiple R-squared:  0.4591, Adjusted R-squared:  0.3464
## F-statistic: 4.074 on 5 and 24 DF, p-value: 0.008074
```

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1(\text{fresh_price}_i) + \hat{\beta}_2(\text{ads_campaignB}_i) + \hat{\beta}_3(\text{ads_campaignC}_i) + \hat{\beta}_4(\text{fresh_price}_i \times \text{ads_campaignB}_i) + \hat{\beta}_5(\text{fresh_price}_i \times \text{ads_campaignC}_i)$$

Interpreting interaction terms

β_0 (intercept) On average, the demand for Fresh is $(12.226 \times 100000) = 1222600$ bottles when fresh price takes on a value of zero and the advertising campaign is campaign A.

β_1 (fresh_price) On average, the demand for Fresh decreases by $(1.079 \times 100000) = 108900$ bottles for each additional R10 increase in the price of Fresh when advertising campaign is campaign A.

β_2 (ads_campaignB) On average, the demand for Fresh is $(35.186 \times 100000) = 3518600$ bottles higher for a company using advertising campaign B compared to a company using advertising campaign A, when fresh_price is R0.

β_3 (ads_campaignC) On average, the demand for Fresh is $(13.538 \times 100000) = 1353800$ bottles higher for a company using advertising campaign C compared to a company using advertising campaign A, when fresh_price is R0.

β_4 (fresh_price*ads_campaignB) The adjustment to the slope for fresh_price when advertising campaign is campaign B compared to campaign A.

β_4 (fresh_price*ads_campaignC) The adjustment to the slope for fresh_price when advertising campaign is campaign C compared to campaign A.

$$\hat{y}_i = 12.226 - 1.079(\text{fresh_price}_i) + 35.186(\text{ads_campaignB}_i) + 13.538(\text{ads_campaignC}_i) - 9.307(\text{fresh_price}_i \times \text{ads_campaignB}_i) - 3.549(\text{fresh_price}_i \times \text{ads_campaignC}_i)$$

When advertising campaign is A:

$$\hat{y}_i = 12.226 - 1.079(\text{fresh_price}_i)$$

When advertising campaign is B:

$$\hat{y}_i = 12.226 - 1.079(\text{fresh_price}_i) + 35.186(1) - 9.307(\text{fresh_price}_i \times 1)$$

$$\hat{y}_i = 47.412 - 10.286(\text{fresh_price}_i)$$

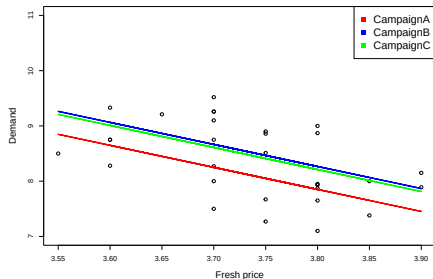
When advertising campaign is C:

$$\hat{y}_i = 12.226 - 1.079(\text{fresh_price}_i) + 13.538(1) - 3.549(\text{fresh_price}_i \times 1)$$

$$\hat{y}_i = 25.764 - 4.628(\text{fresh_price}_i)$$

Additive model:

$$\hat{y}_i = \beta_0 + \beta_1(\text{fresh_price}_i) + \beta_2(\text{ads_campaignB}_i) + \beta_3(\text{ads_campaignC}_i)$$



Multiplicative model:

$$\begin{aligned} \hat{y}_i = & \beta_0 + \beta_1(\text{fresh_price}_i) + \beta_2(\text{ads_campaignB}_i) + \\ & \beta_3(\text{ads_campaignC}_i) + \\ & \beta_4(\text{fresh_price}_i \times \text{ads_campaignB}_i) + \\ & \beta_5(\text{fresh_price}_i \times \text{ads_campaignC}_i) \end{aligned}$$

